In RAG We Trust?
Ensemble Models for Retrieval Augmented Generation

Christian R. Cuenca
Dipartimento di Informatica
Università degli Studi di Milano, Italy

Introduction

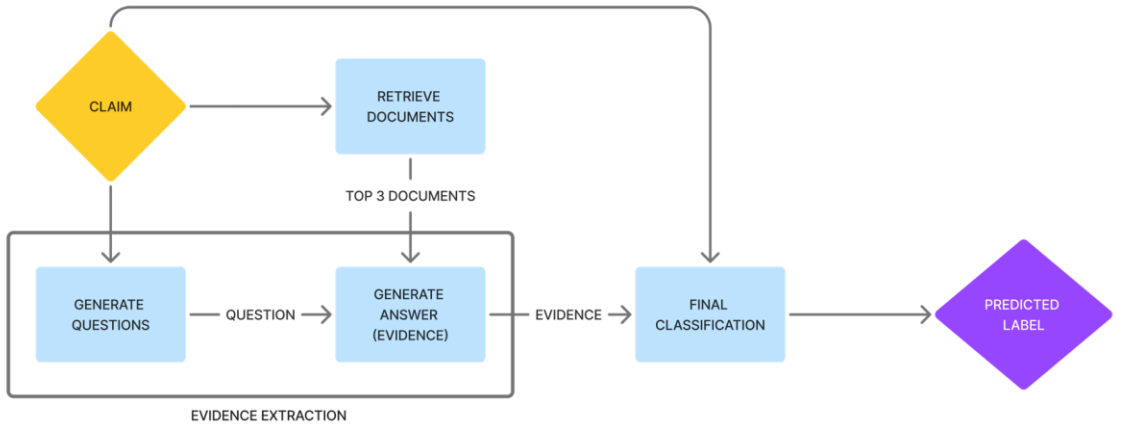
In recent years, **information management** has become increasingly critical.

On social media, **fake news and misinformation** can lead to real-world harm.
The risk is amplified when topics involve **elections, pandemics, or other sensitive issues** [1,2]

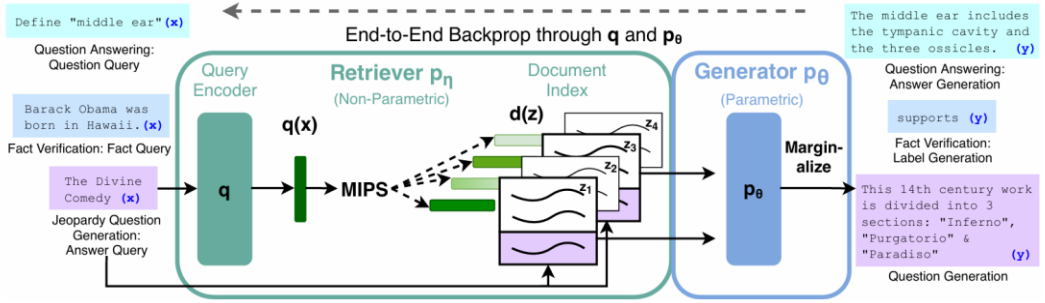
In Enterprise settings is also particularly critical since **errors in information** may lead to **operational or regulatory consequences**, for example in Compliance, Governance, Security policies, Internal procedures.

Several countermeasures have been proposed in the literature. For example:

Singhal et al. [3] proposed an evidence-backed fact verification pipeline that classifies claims into four categories: Supported, Refuted, Conflicting Evidence/Cherrypicking, and Not Enough Evidence



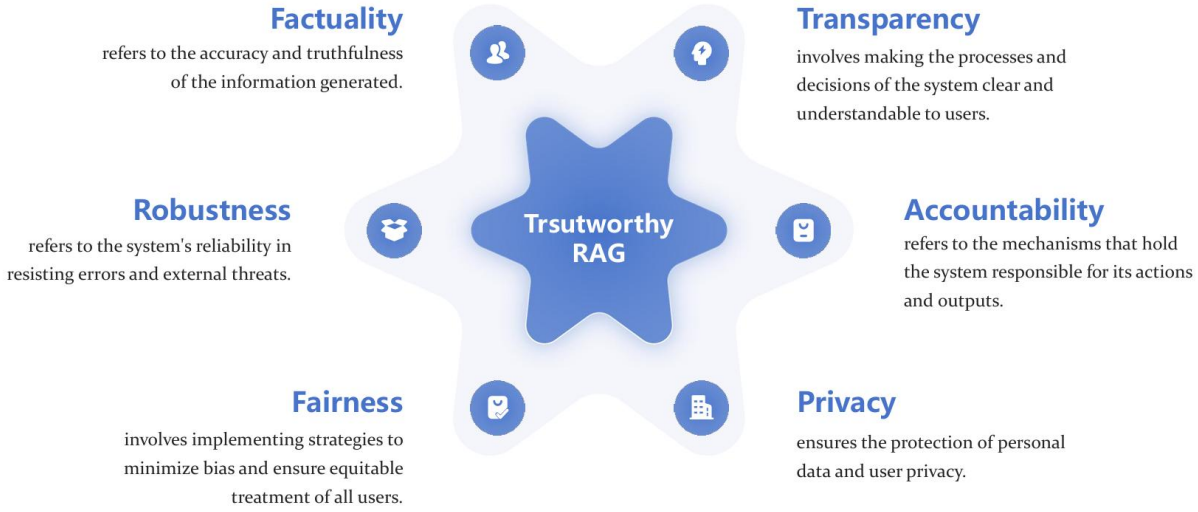
Introduction



Lewis et al.[4] introduced Retrieval-Augmented Generation (RAG), which a **generative model retrieves relevant documents** from an **external index** (e.g., Wikipedia) before or during answer generation. This approach addresses **two key limitations** of standalone LLMs:

- model **knowledge is encoded in parameters and may become outdated** without retraining;
- generated answers can be grounded in explicit, **traceable sources.**

Despite these advantages, RAG pipelines are **not immune to errors** and may still generate inaccurate or misleading outputs. Zhou et al. [5] propose a benchmark framework to **assess trustworthiness across six key dimensions:**

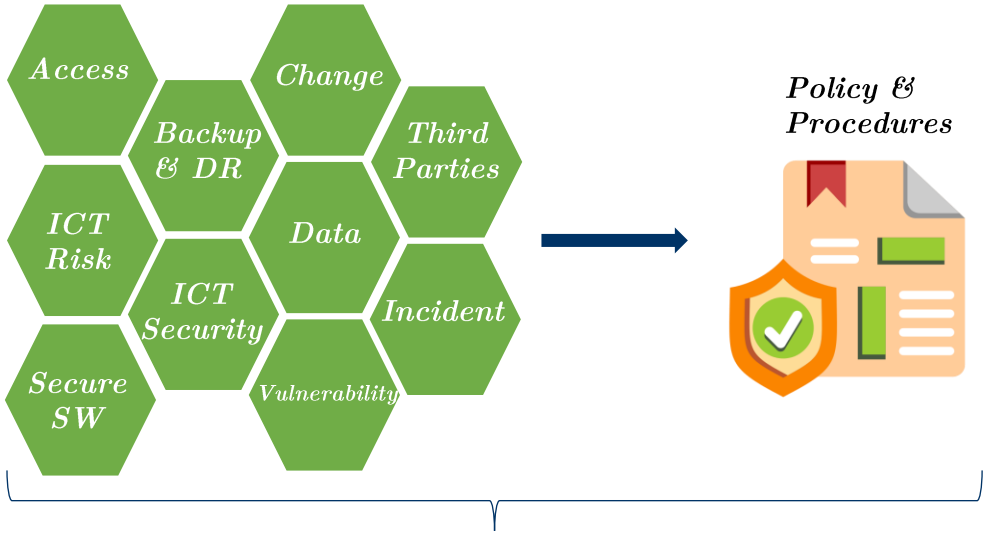


Materials and Methods

In this work, we evaluate the behavior of a RAG system, specifically a RAG architecture based on an ensemble of generative models.

Rather than **replicating the experimental setups proposed in prior studies**, we adopt a more **practical perspective, grounded in an enterprise setting**.

We simulate a realistic **cybersecurity governance scenario** and, within this context, analyze the system's behavior in the presence of **retrieval poisoning** affecting the underlying information base.

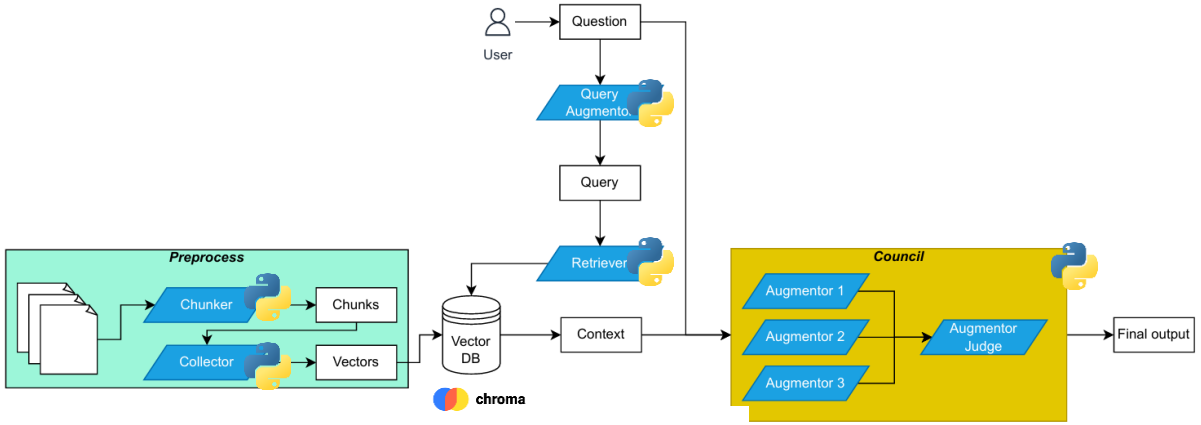


Metrics

- 10 textual documents
- Synthetic data generated using a commercial LLM (OpenAI ChatGPT 5.2)
- Stored in *.txt* to simplify preprocessing (in real settings, documents would typically be in another format)
- Average length: 984.3 words per document
- Longest document: 1865 words
- Shortest document: 323 words
- Each document is structured into numbered sections and subsections

Materials and Methods

The pipeline presented in this work begins with a preprocessing stage. The inference phase is triggered by a user question, which is then passed to a Council of LLMs* for deliberation.



Preprocess

- The .txt documents are processed by chunker.py, which splits each document into section-level .txt files (facilitating data visualization)
- A total of **204 chunks** are generated
- The collector.py module encodes each chunk into dense vectors using the **all-MiniLM-L6-v2** model
- Embeddings are computed using only the **section body text**, excluding section numbers and titles
- Vectors are stored within a **persistent ChromaDB collection**. An HNSW (Hierarchical Navigable Small World) index is built over the stored vectors to enable efficient approximate nearest neighbor (ANN) search during retrieval

Inference

- A **natural-language question** is first processed by an LLM-based Query Augmentor, instructed to reformulate it into more **retrieval-efficient queries**.
  - Example: "How does the identity lifecycle management process work from joiner to leaver?"→ "identity lifecycle management process joiner leaver"
- The Retriever encodes the enhanced query, performs similarity search over the vector database using **cosine similarity**, and **returns the top-*k*** most relevant chunks. These are concatenated into a single text block, forming the context provided to the generative models.

Council

- The retrieved context block is provided to multiple LLMs, each prompted with a **specific expert role** (as a conservative banking compliance expert, a pragmatic ICT operations engineer, and a critical security auditor)
  - Example: You are {role prompt} Answer using the context below. Context: {context} Question: {question}
- A final **Judge model receives** the same inputs (question and context) along with the candidate answers, resolves potential conflicts, and produces a **single response** returned by the system

*inspired by [6]

Experiment - Factuality

In this experiment, we investigate scenarios where the model's internal (parametric) knowledge conflicts with the external evidence retrieved from the document corpus.

Strategies

- Role-based models: **OpenAI ChatGPT 4.1 nano**
- Retrieval configured with **top-k = 1** (to intentionally stresses the system by exposing it to cases where the single retrieved document may be incomplete or weakly relevant)
- Constructed **50 questions**, each annotated with the expected source document
- Added **5 out-of-corpus questions** to evaluate system behavior when no relevant information is available in the knowledge base

Council prompt

You are {role prompt}
Answer using also the context below. Be concise.
Context: {context}
Question: {question}

Output in this format:
Answer: <your brief answer>
Used Model's Internal Knowledge: <YES | NO >

Result

- Results were carefully **reviewed by a domain expert**
- Retrieval outcomes were **annotated into three categories**:
  - YES – retrieved document matches the expected source
  - YES BUT – not the expected document, but still relevant
  - NO – retrieved document does not match the expected source

Metrics

Exact match

$$EM@1 = \frac{|YES|}{|Questions|} = 0.38$$

Acceptable

$$\text{Acc@1} = \frac{|YES| + |YES\ BUT|}{|Questions|} = 0.68$$

Irrelevant

$$\text{Err@1} = \frac{|NO|}{|Questions|} = 0.32$$

Off Target But Relevant

$$\text{OTBR} = \frac{|YES\ BUT|}{|Questions|} = 0.30$$

Judge prompt

Question: {question} Context: {context}
Answers from multiple experts:{multi-answers}

Task:- Compare the answers
- Resolve contradictions
- - Produce a single, accurate, well-justified final answer
- - Be concise
- - Provide the answer with the following format:
- ***Final Answer***: <your BRIEF final answer here>
- ***Used Model's Internal Knowledge***: < YES | NO >

Experiment - Poisoning

In this experiment, we investigate scenarios where the system is under adversarial condition, i.e. retrieval poisoning.

Strategies

- Role-based models: **OpenAI ChatGPT 4.1 nano**
- For each question in the test set, we **replaced the retrieved evidence** with passages that are **topically relevant but factually incorrect**
- **Poisoning strategies**:
  - Numerical flips (e.g., SLA timelines, retention periods, thresholds; "within X days → Y days")
  - Role swaps (e.g., inverting responsibilities across HR, IAM, and the CISO)
  - Control inversion (e.g., "MUST encrypt" → "encryption optional")
  - Exception injection (e.g., "for urgent cases, approvals can be bypassed")
- Varied **poisoning intensity**:
  - Fully poisoned
  - Partially poisoned
  - Poisoned with Contraindication
  - Unchanged

Out of 50 questions, 43 were poisoned in a subtle manner (where 8 left unchanged), 7 were poisoned in an obvious manner

Results

- Measured **how often the model explicitly flags** the provided evidence as fully or partially **poisoned**.
- Across the 50 questions, the system identified 1 case as fully poisoned and 1 case as partially poisoned

Council prompt

```
You are {role prompt}
Answer using also the context below. Be concise.
Context: {context}
Question: {question}

Output in this format:
***Answer***: <your brief answer>
***Used Model's Internal Knowledge***:  <YES | NO >
***Poisoned***: <FULL|PARTIALLY|NO>
```

Judge prompt

```
Question: {question} Context: {context}
Answers from multiple experts:{multi-answers}

Task:- Compare the answers
- Resolve contradictions
- - Produce a single, accurate, well-justified final answer
- - Be concise
- - Provide the answer with the following format:
- ***Final Answer***: <your BRIEF final answer here>
-  ***Used Model's Internal Knowledge***: < YES | NO >
- ***Poisoned***: <FULL|PARTIALLY|NO>
```

Consideration (1/2)

In this work, we assess the factuality dimension of RAG trustworthiness following the methodology proposed by Zhou et al. [5]

1	To stress-test factuality we constrained the retrieval by setting top-k=1	With larger k, the system would eventually retrieve at least one truly relevant passage and thus answer correctly more often, masking failure modes that emerge under weak or noisy evidence.
2	The retriever's performance (e.g., EM@1) is also influenced by our preprocessing design	Chunks are embedded using only the section body text, excluding section titles (for semantically cleaner representations). But it also introduces cases where the most similar chunk is not the expected one, despite being partially informative. "Describe the ICT risk governance structure and responsibilities" the expected evidence is the roles and responsibilities section of *ICT Risk Management Policy.txt*, but instead the retriever returns the introduction section, which still contains generic but relevant governance information.
3	When retrieval is not sufficiently aligned with the question, the system fall back on parametric knowledge despite being instructed to rely only on the provided context	Reliance on parametric knowledge is not uniform across the Council: in some instances, only some Augmentors uses its own knowledge. In the final output, the Judge reports parametric knowledge usage only when it reflects the majority of the Council (or the Judge's own assessment)

Consideration (2/2)

In this work, we assess the factuality dimension of RAG trustworthiness following the methodology proposed by Zhou et al. [5]

4	For the intentionally out-of-corpus questions, the system generally behaved appropriately	Either stated that the answer could not be found in the provided documents, or it responded using internal knowledge. "Which exact ISO 27001 controls (Annex A) are adopted as mandatory, with control IDs?" → the model produced a correct answer based on its background knowledge of ISO 27001.
5	Regarding retrieval poisoning, the system flagged poisoning only twice	One case as fully poisoned (in obvious manner) and one as partially poisoned (in subtle manner) In 4 instances where at least one Augmentor raised a poisoning concern; However, these signals were not retained in the final adjudication, suggesting that the Judge step can suppress minority warnings even under the current prompting strategy.
6	Poisoning effectiveness depends on question–poison alignment	For instance, a role-swap injection (e.g., incorrectly assigning backup responsibilities to HR) has limited impact if the question targets implementation standards rather than accountability, such as "What implementation standards are required for backups?"

Conclusion

Overall, the results support the practical value of RAG in enterprise contexts. In document-centric workflows, such as gap analysis and audit, RAG can simplify evidence access and synthesis, and an ensemble-based design can further improve reliability.

Two practical mitigation directions emerge:

1. Increasing top-k can help “dilute” compromised evidence by providing additional passages that may counterbalance poisoned ones.
2. The prompting and decision rule of the Council/Judge could be tuned to adopt a more conservative aggregation strategy (e.g., treating any poisoning flag raised by a Council member as a positive signal) an approach often favored in security settings where false positives can be preferable to false negatives.

As future work, it would be valuable to study vector level poisoning more directly, quantifying how often poisoned passages are retrieved and under which embedding/index con figurations this becomes more likely

Also considering the multi LLM approach might be expensive in some case, it would be valuable to implement a small local LLM where possible

Thank You

Bibliography

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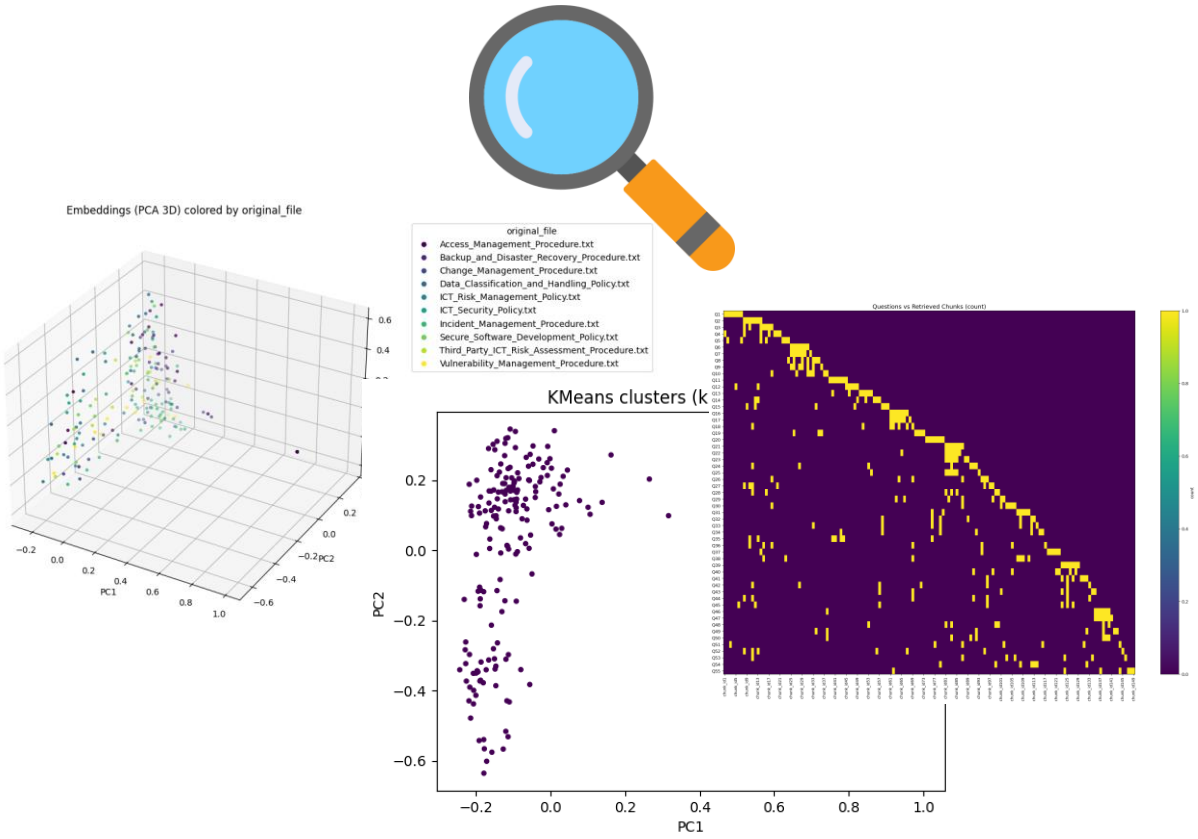
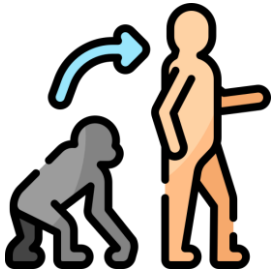
Annex

Some Supporting materials are provided below, including selected data visualizations.
Please refer to the **Jupyter notebook** for data visualization details and to the **test_sets.xlsx** file for the complete experimental results, present on the GitHub repository





The GitHub repository documents the evolution of the project across different branches, showing the progression from a basic RAG implementation to the final Council-based architecture:

1. Naive RAG – Initial implementation with a standard retrieval–generation pipeline.
2. Stateful RAG – Introduction of conversational memory, enabling stateful interactions across user turns.
3. Query Augmentor – Integration of a query-enhancement module to improve retrieval effectiveness.
4. Council Architecture – Adoption of a multi-model ensemble with role-specialized LLMs and a final Judge for consolidated decision-making.